

• Basic Parabolic Equation

$$\begin{cases} \partial_t u - \Delta u = f & \text{in } \Omega \times (0, T) & \text{space} \times \text{time} \\ u = 0 & \text{in } \partial\Omega \times (0, T) & \text{boundary condition} \\ u(\cdot, 0) = u_0 & \text{in } \Omega & \text{initial condition} \end{cases}$$

Variational form: (weak form)

$$(\partial_t u, v) + a(u, v) = (f, v) \quad \forall v \in H_0^1(\Omega)$$

$$\begin{array}{c} \uparrow \qquad \qquad \qquad \uparrow \\ \text{ii} \\ (\nabla u, \nabla v) \end{array}$$

2 duality pair options: $\begin{cases} L^2 \text{ inner product } (\cdot, \cdot) \\ H^1(\Omega) \text{ and } H_0^1(\Omega) \langle \cdot, \cdot \rangle \end{cases}$

the solution should be $[u, \partial_t u]$.

- FEM semi-discretization in space

why called semi-discretization? Because in this step, we see time t fixed, and then discretize w.r.t spatial variable on Ω . The solution from this process is still continuous in terms of time t .

T_h is triangulation of Ω .

V_h is finite element space based on T_h

$$\text{P.D.E: } \begin{cases} (\partial_t u_h, v_h) + a(u_h, v_h) = (f, v_h) & \forall v_h \in V_h \\ u_h(\cdot, 0) = u_{0,h} \end{cases}$$

* only show initial condition as $V_h \subset H_0^1(\Omega)$ already indicates boundary condition.

Problem: given $f \in V_h' \times (0, T]$, $u_{0,h} \in V_h \subset H_0^1(\Omega)$
WTF $u_h \in L^2(0, T; V_h)$

e.g. V_h is linear finite element space.

let φ_i be standard hat basis at vertex x_i , $i=1, \dots, N$

expand our u_h :

$$u_h = \sum_{i=1}^N u_i(t) \varphi_i(x)$$

so $(\partial_t u_h, v_h) \xrightarrow[\text{test by basis } \varphi_j]{\text{break down}} \left(\sum_{i=1}^N u_i'(t) \varphi_i(x), \varphi_j(x) \right)$

$\uparrow$
interiors

generally,
as v_h arbitrary,
should be tested by all basis

$$u_j'(t) \|\varphi_j\| = u_j'(t)$$

$\uparrow$
this is using 3-vertices
quadrature to achieve
mass-lumping

i.e. diagonalize mass matrix M

$$\begin{cases} \sum_{i=1}^N (u_i'(t) \varphi_i(x), \varphi_1(x)) \\ \sum_{i=1}^N (u_i'(t) \varphi_i(x), \varphi_2(x)) \\ \vdots \\ \sum_{i=1}^N (u_i'(t) \varphi_i(x), \varphi_N(x)) \end{cases}$$

$$\begin{bmatrix} (\varphi_1, \varphi_1) & (\varphi_1, \varphi_2) & \dots & (\varphi_1, \varphi_N) \\ (\varphi_2, \varphi_1) & \dots & & \\ \vdots & & & \\ (\varphi_N, \varphi_1) & & & (\varphi_N, \varphi_N) \end{bmatrix} \begin{bmatrix} u_1'(t) \\ u_2'(t) \\ \vdots \\ u_N'(t) \end{bmatrix}$$

$$(\varphi_i, \varphi_j) = 0 \text{ if } i \neq j$$

$\uparrow$ if we used 3 vertices
quadrature, then

$$M = \text{diag}(\varphi_i, \varphi_i)$$

$M u$

M : mass matrix

same process. $Q(u_h, v_h) \xrightarrow[\text{test by basis } \varphi_j]{\text{break down}} Q(\sum_{i=1}^N u_i \varphi_i(x), \varphi_j(x))$

make use of the property of bilinear form.

similar to preceding $M = [(\varphi_i, \varphi_j)]_{N \times N}$, $M u$

this is $A = [a(\varphi_i, \varphi_j)]_{N \times N} = [(\nabla \varphi_i, \nabla \varphi_j)]_{N \times N}$, $A u$.

A : stiffness matrix

RHS $\langle f, v_h \rangle \xrightarrow[\text{as corresponding coordinate}]{\text{test by basis } \varphi_j, \text{ break down}} \langle f, \varphi_j \rangle$

~~consider $\int f(x) dx \approx \frac{1}{3} \sum_{i=1}^3 f(x_i) |T|$~~

intro above are all general cases.

when it turns to 2-D case:

mass matrix $M = \text{diag}(m_i)$ as $m_i = h^2$
 stiffness matrix A is the same as 5-point stencil FDM
 RHS consider $\int f(x) dx \approx \frac{1}{3} \sum_{i=1}^3 f(x_i) |T|$

all could be verified in details if needed to code.

- FEM semi-discretization in time

the simple idea is just dividing time interval $(0, T)$ into uniform grid with gap size $\delta t = \frac{T}{N}$. and we have $t^n = n \cdot \delta t$ with $n = 0, 1, 2, \dots, N$

① Forward Euler Method :

$$\begin{cases} \frac{u^n - u^{n-1}}{\delta t} + Au^{n-1} = f^{n-1} \\ u^0 = u_0 \end{cases}$$

$\delta t \leq h^2$ stability requirement for forward Euler method.

- FEM semi-discretization in space

$$\underbrace{Mu}_{\partial_t} + \underbrace{Au}_{\Delta} = f \quad (\text{must in weak form})$$

↑
using integral quadrature